James Ensminger

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EXPERIENCE

Applications Engineer Nov 2023 - Present
PowerFlex | Mountain View, CA and Los Angeles, CA
- Enhanced EV charger communication reliability through bench testing, innovative tooling, and data-driven R&D.
- Advanced distributed control system design iterations with the HW team for optimized power distribution, networking, and telemetry between onsite network devices and DERs.
- Led the development and launch of customer-facing and internal-facing technical documentation platforms on PowerFlex HW and SW products and services.

Asset Health and Performance Center Engineering Intern Jun 2021 - Sep 2021
Pacific Gas and Electric | San Francisco, CA
- Leveraged GIS to provide informative data on future EFD (Early Fault Detection) and DFA (Distribution Fault Anticipation) line sensor technology installments across Northern California's electric grid.
- Managed and proactively coordinated with multiple teams to compile a quarterly voltage curtailment report on all solar powered line sensor units, while taking initiative to reduce data compilation processing time by 20% (saving 1 hour) through root cause analysis and process improvement methodologies.

PROJECTS AND ACHIEVEMENTS

3x3 Systolic Array Mar 2026 - May 2026
- Fully custom design in 45nm CMOS for accelerating matrix multiplication by enabling parallel computation across 9 synchronous PEs (processing elements) each performing MAC (multiply-accumulate) operations.
- Achieved 385MHz maximum operating frequency, 450μW average power, and 21,730μm² layout footprint with clean DRC/LVS sign-off.

Multi-Device Wireless Charger Dec 2022 - Jun 2023
- Created a Qi standard wireless charger with increased freedom of placement for multi-device charging, an intuitive visual charge indicator determined by stakeholders, and an efficient heat management system enabling it to operate 5°C cooler than chargers on the market.

Ground Bounce PCB Test Circuit Mar 2022 - Jun 2022
- Designed a PCB test circuit to experimentally observe the electrical phenomena of ground bounce.
- Developed the PCB design with OrCAD Capture and Allegro PCB Editor, then milled it using an M60 LPKF.

EDUCATION

University of Southern California | Master of Science in Electrical Engineering Aug 2025 - Present
- **Research:** Flexible Electronics at USC Center for Advanced Manufacturing
- **Coursework:** Modern Solid-State Devices, Power Systems, MOS VLSI Design, Nanotechnology, Computing Principles, VLSI System Design

University of California, Santa Cruz | Bachelor of Science in Electrical Engineering Sep 2019 - Jun 2023
- **Clubs:** MEP, NeuroTechSC, SlugSat, UCSC IEEE
- **Grader:** ECE178 and ECE224 Semiconductor Device Electronics [Winter Quarter 2023]

SKILLS, TOOLS, AND TECHNOLOGIES

- **Electrical:** OrCAD Capture, PSpice, Allegro PCB Editor, Altium Designer, Cadence Virtuoso, Ansys Maxwell, Hardware Development/Debugging, Microcontrollers, RTL Design, FPGA, PLC programming (IEC 61131-3), Modbus TCP, Computer Networking, Soldering, Lab Equipment (Oscilloscopes, Signal Generators, Power Supplies, DMMs, Spectrum Analyzers)
- **Software:** Verilog, Python, C/C++, Java, SQL, PowerShell, Linux, Git, MATLAB
- **Mechanical:** SolidWorks, 3D Printing, Ansys Mechanical